

Demo: A Lightweight Emulation Framework for Energy-Aware Federated Learning

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ABSTRACT

IoT networks face critical challenges in energy efficiency, privacy, and communication reliability. Federated Learning (FL) enables collaborative model training without sharing raw data, but traditional client selection can drain energy and disrupt RPL-based networks. This demo introduces MininetFed, a network emulator with FL support in an energy-aware client selection strategy. Clients are chosen based on energy availability, balancing model accuracy and network longevity. Users can interact with different topologies and selection algorithms, with results showing enhanced learning efficiency and extended network lifetime in multi-hop IoT scenarios.

CCS CONCEPTS

• **Networks** → **Network experimentation**;

KEYWORDS

Federated Learning, Internet of Things, MininetFed

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1 INTRODUCTION

According to Gartner¹, the global Internet of Things (IoT) market is projected to reach \$991 billion by 2028, with over 75 billion connected devices. More than 95% of new industrial IoT deployments are expected to integrate analytics and Artificial Intelligence (AI) capabilities. As the number of connected devices continues to rise, AI-driven IoT networks are becoming increasingly complex. Therefore, optimizing resource management, energy efficiency, and communication in rapidly growing sectors, such as manufacturing,

transportation, oil and gas, environmental monitoring, automotive, and healthcare, has become essential.

FL has emerged as one of the most promising AI-based solutions for IoT networks and applications. FL enables collaborative model training without the need to share raw data between devices, significantly reducing communication and storage costs while ensuring a high level of privacy for devices/users. FL can use client selection strategies to improve communication overhead, energy efficiency, and accuracy in heterogeneous IoT scenarios.

Given the inherent costs and complexities of IoT FL environments, it is crucial for researchers and practitioners in the networking industry to implement and evaluate their proposals by using simulation and emulation tools, such as TensorFlow Federated and Flower, to better understand and improve IoT application scenarios and algorithms [6]. To address this need, emulation platforms are key solutions in designing and evaluating FL systems across IoT networks without the need for real-world implementation. Realistic emulation capabilities offer substantial technical and financial benefits, enabling researchers and practitioners to accelerate the market readiness of their innovative solutions, while minimizing risks and reducing operational costs.

This demonstration showcases MininetFed [5], an emulation platform that combines energy-aware monitoring with intelligent client selection mechanisms for FL. By favoring energy-efficient nodes, MininetFed reduces network disruption and sustains high global model accuracy. MininetFed offers a platform for evaluating FL-based IoT use cases at scale, helping researchers and industries reduce costs, validate solutions before deployment, and improve system efficiency.

2 SYSTEM DESIGN

MininetFed² extends Mininet-WiFi [2] and Containernet [4] to emulate realistic FL environments, allowing users to configure heterogeneous devices in terms of CPU, memory, and network connectivity. Among its key features, MininetFed allows users to:

- Define data distribution strategies across devices
- Implement custom client selection policies
- Choose aggregation functions for model updates
- Utilize pre-trained models for well-known datasets
- Visualize key performance metrics through graphical representations

MininetFed emulates device communication using WiFi, IEEE 802.15.4, and wired interfaces, providing a flexible and realistic

¹www.gartner.com/en/documents/5499495

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²<https://github.com/lprm-ufes/MininetFed>

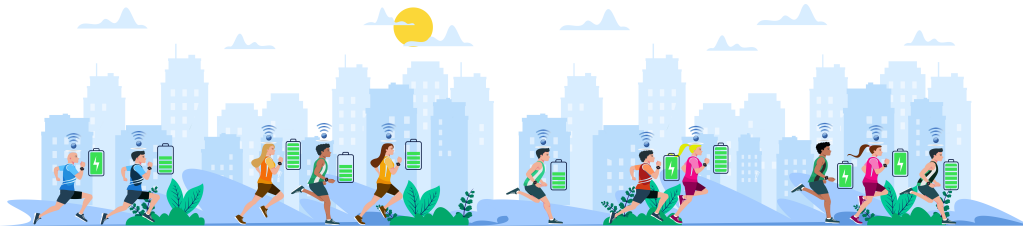


Figure 1: Street Race Scenario.

environment for heterogeneous IoT networks. It also supports P4-programmable data planes, enabling advanced experimentation with custom packet processing and in-network computing tailored to FL scenarios. Through a single configuration file, users can define a wide range of hardware characteristics for each container that emulates an IoT device, including memory, CPU, interface types, and battery capacity, allowing fine-grained control over resource constraints and system behavior. This modular setup facilitates reproducible experiments and rapid prototyping of FL strategies under diverse network conditions and device capabilities.

The architecture consists of at least one aggregation server, a group of client nodes, and an MQTT (Message Queuing Telemetry Transport) broker to handle communication. In contrast to frameworks like Flower [1], which rely on synchronous gRPC-based communication between clients and servers. By adopting an asynchronous communication model based on the publish-subscribe paradigm, MininetFed enables the server to disseminate global model updates and receive client contributions independently. This approach enhances scalability and flexibility, particularly in dynamic and resource-constrained environments.

3 DEMONSTRATION USE CASE

Monitoring heart beats per minute (BPM) during physical exercise is critical to ensuring athlete safety. However, BPM sensors are often energy-intensive [3]. As a result, these devices commonly disable BPM sensor and estimate BPM values based on demographic attributes (e.g., skin tone, age, weight, height) and low-cost sensor data, such as temperature, gyroscope, and accelerometer. These estimations, however, tend to be inaccurate as they rely on embedded predictive models. Federated Learning addresses this challenge by training a BPM estimator model locally on each user device, sharing only the model parameters with a central server. This approach improves estimation accuracy while safeguarding sensitive user data and preserving privacy.

The presented demo showcases the MininetFed's capabilities by emulating a street race scenario, such as a marathon, with participants wearing smartwatches and collaboratively improving a BPM prediction model using only low-cost and energy-efficient sensors. The demo focuses on a race segment where a data collection point (e.g. Base Station) receives locally trained models from athletes, performs model aggregation, and distributes the updated global model back to devices, as illustrated in Figure 1. Communication among devices and the aggregator occurs over an LoWPAN RPL-based network with a maximum but not limited of five hops.

Demonstration steps: Participants will have the opportunity to interact with the MininetFed platform, exploring various configurations in a emulated IoT environment. They can select among at least two different client selection algorithms-such as selection

of all clients and a random baseline-within a street race scenario. Furthermore, users will be able to modify the network topology and adjust the number of participating nodes, enabling real-time experimentation with different FL scenarios and configurations.

Preliminary results: MininetFed effectively emulates IoT-based networks with integrated FL support across diverse topologies and settings. During the demonstration, a real-time graphical interface will visualize key performance indicators, offering insights into the behavior and impact of different client selection strategies on model accuracy, energy consumption, and overall network performance.

4 CONCLUSIONS

This demonstration introduces MininetFed, an emulation platform designed for experimenting with energy-aware Federated Learning in Internet of Things networks. It enables users to explore diverse network topologies, node configurations, and client selection algorithms, while real-time visual analytics provide actionable insights into their impact on system performance and energy efficiency.

In future work, MininetFed will add support for heterogeneous IoT devices, such as UAVs, and optimize modeling in resource-constrained environments. Additionally, mobility support will be integrated, enabling realistic wireless channel emulation for dynamic topologies and mobile nodes. This enhancement will allow for the emulation of critical scenarios that require mobility, such as drone-based environmental monitoring, vehicular networks, smart transportation systems, and disaster response operations, where devices frequently change position and connectivity patterns.

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